# Improving Cantonese translation in existing multilingual translation models

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## Abstract

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## 1 Related works

### 1.1 MADLAD

MADLAD As a base for our experiments, and later as a model to finetune, we use the MADLAD model. This is a general translation model, with 10.7B parameters, trained on web-scraped unlabeled data in 419 languages. The dataset contains both Mandarin and Cantonese data, but the amount of Cantonese data is very limited compared to the amount of Mandarin data (approximately 61 times less). The model architecture is a standard transformer-based sequence-to-sequence model, similar to the architecture used in models such as T5[**?** ].

## 2 Introduction

This project explores improving machine translation tasks working with Cantonese text. Although Cantonese has more than 85 million speakers, it is very underrepresented in machine learning and translation, as large training corpusses are rare as the language is most used in its spoken form. This results in many Cantonese NLP tasks simply being handled by models trained mostly or exclusively on Mandarin language data.

In this project, we explore how both unsupervised and supervised learning can be applied to improve the quality of translation tasks. For this, we use the professionally translated data in the SMOL[1] and GATITOS[3] datasets and the unlabeled data in the dataset collected by Dare et al.[2].

The first experiment performed was a simple token replacement. As there is overlap between Cantonese and Mandarin characters, we replace the tokens that are different in the Cantonese input with their Mandarin equivalents before feeding the text to a pre-trained MADLAD[5]. This simple approach showed some promise, but was limited by not being able to handle differences in grammar and differences in meaning of characters that are the same in both languages.

We use the unlabeled data to train Cantonese token embeddings and then transform these embeddings into the embedding space of a pre-trained English-Mandarin model, MADLAD, using the token-level translations in GATITOS. This TODO: Results of this.

We then finetune MADLAD on the SMOL dataset and evaluate the performance of the finetuned model on a held-out test set from SMOL. TODO: Results of this.

# 3 Datasets

## 3.1 SMOL

SMOL[1] (Set of Maximal Leverage) is a high-quality translation corpus from English to more than a hundred low-resource languages. SMOL includes SMOLSent and SMOLDoc, which represent sentence-level and document-level translations is nicely accompanied by Gatitos[3] which provides token-level translations across many of the same languages. In this project specifically, we use the English-Cantonese portion of SMOL and GATITOS, where there are translations on all three levels of data.

## 3.2 Monolingual Cantonese corpus by M. Dare, et. al

In their work on unsupervised Mandarin-Cantonese machine translation, Dare et. al.[2] collected a large corpus of monolingual Cantonese sentences from various online sources. In total 912258 lines were collected by scraping and cleaning data from Wikipedia, Instagram, and a few other sites including a few corpusses compiled for previous academic work. Common for most sources that form the corpus is that the content is mostly user-generated text, which often includes non-standard spellings and colloquial language, which is typical for Cantonese as a written language. We use this corpus to train Cantonese token embeddings for the embedding transform experiment described in Section 3.

# 4 Baseline

We evaluate our model on a subset of the SMOLSent dataset using BLEURT.

# 5 Experiments

## 5.1 Token replacement

Since the MADLAD model appears to have difficulty with sentences that contain words that are native to written Cantonese, we do an experiment where we replace words that are native to written Cantonese with their Mandarin Chinese translations. To find words that are native to written Cantonese we use CantoDict[7], a collaborative project where Cantonese speakers annotate Cantonese characters with information about their meaning, pronounciation and usage in written Cantonese and Mandarin. If a character in a sentence is used in Cantonese but not used in Mandarin, we try to find its translation in the Gatitos dataset and do a token-level replacement as preprocessing to MADLAD inference.

## 5.2 Embedding transform

From the promising results of the token replacement experiment 6.1, we believe that giving the model more knowledge of Cantonese characters could significantly improve its Cantonese translation ability.

If we were to simply train a model on Gatitos token-level data it might get slightly better at some characters in the small Gatitos dataset, but we would not be able to generalize the model to become better at a significant amount of Cantonese vocabulary.

Instead we propose the following experiment: We train embeddings on a large corpus of unlabeled Cantonese data, hoping to get embeddings that better represent semantic similarity between

Cantonese tokens than the embeddings of MADLAD do. We then perform an embedding transform from our Cantonese embeddings to MADLAD's Chinese embedding space using token-level translations from Gatitos. Specifically, we train a transformation matrix $W$ such that for a specific Cantonese-Chinese token embedding pair $E_{canto}, E_{mandarin}$ the embedding transformation $E_{canto}W$ approximates $E_{mandarin}$. This training was simply done optimizing the loss function $||E_{canto}W - E_{mandarin}||$ with the ADAM-algorithm[4].

For training the Cantonese embeddings we use the Monolingual Cantonese corpus by M. Dare, et. al. to train a Word2Vec[6] model.

### 5.3 SMOL finetuning

We do back-translation finetuning for the MADLAD model on the SMOLSent and SMOLDoc English-Cantonese datasets. We flatten the SMOLDoc dataset such that it consists of sentence pairs like SMOLSent, and combine the two datasets using a 80/10/10 split for training/validation/test.

## 6 Results

### 6.1 Token replacement

### 6.2 Embedding transform

### 6.3 SMOL finetuning

We finetune with a learning rate of ??? and a batch size of $8$ with a weight decay of $0.01$ using a A100 GPU on Google Colab. After $1$ epoch ($784$ steps) the model starts to overfit as can be seen on figure TODO: Insert figure and figure reference, so we use a checkpoint of the model saved after the first epoch.

## 7 Discussion

### 7.1 Limitations

In the embedding transform experiment we may have achieved better results if we had pretrained a T5 model on our embeddings using masked sentence modelling, like MADLAD had used.

### 7.2 Other methods

Non-linear transformations of the embeddings, such as applying ReLU or sigmoid activation functions... TODO: Why we did not try this, too likely to overfit given the size of Gatitos

## References


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